Attention Is All You Need: Transformer Architecture Overview

1. Introduction to the Transformer
The Transformer model, introduced by Vaswani et al. in 2017, completely replaces recurrent neural networks (RNNs) and convolutional neural networks (CNNs) with self-attention mechanisms. Unlike RNNs that process sequences sequentially token-by-token, Transformers process entire sequences simultaneously, enabling superior parallelization during training.

2. Self-Attention and Scaled Dot-Product Attention
At the core of the Transformer is the Scaled Dot-Product Attention mechanism. Given queries Q, keys K, and values V, the attention score is computed as:
Attention(Q, K, V) = softmax((Q * K^T) / sqrt(d_k)) * V
where d_k is the dimension of the keys. The scaling factor 1/sqrt(d_k) prevents the dot products from growing excessively large for large dimensions, which would otherwise push the softmax function into regions with dangerously small gradients.

3. Multi-Head Attention
Instead of performing a single attention function, Multi-Head Attention linearly projects queries, keys, and values h times into lower-dimensional spaces. This allows the model to jointly attend to information from different representation subspaces at different positions.

4. Positional Encoding
Because the Transformer contains no recurrence and no convolution, it is inherently permutation-invariant. To give the model information about the relative or absolute order of tokens, positional encodings (using sinusoidal functions of different frequencies) are added to the input embeddings at the bottoms of the encoder and decoder stacks.

5. Feed-Forward Networks and Residual Connections
Each layer of the encoder and decoder contains a fully connected feed-forward network applied to each position separately and identically. Residual connections followed by layer normalization (Add & Norm) surround each sub-layer to stabilize gradient flow during deep backpropagation.